

AI-based Elderly Fall Detection Using Pose Estimation and YOLO

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Abstract—Falls pose a serious risk to the elderly, which requires accurate and timely detection to prevent injuries. Existing systems face challenges due to complex environments and limitations of wearable sensors and traditional methods. This work proposes a vision-based fall detection system that uses YOLO to detect persons and beds, combined with AlphaPose and ST-GCN for pose estimation and action recognition, employing a custom logic based on posture and position relative to the bed. The experimental results show that the proposed system provides reliable fall detection in various real-life scenarios, enhancing the safety of at-risk individuals.

Index Terms—Fall Detection, Elderly Care, Computer Vision, YOLOv8, YOLOv11, AlphaPose, ST-GCN.

I. INTRODUCTION

Falls among the elderly have become a major global health concern. The population of 60 years and older is projected to reach almost 2 billion by 2050, about 22% of the world's population [1]. Falls have often caused severe health problems, including loss of function and increased mortality [2], and have remained a leading cause of injuries in older adults [3]. Approximately 28% to 35% of people over 65 have experienced falls yearly, rising to 42% for those over 70 [1].

Fall causes have been multifactorial, involving physiological decline, medical conditions, and environmental hazards [4]. Age-related balance and strength loss, neurological disorders, and poor living conditions have increased risk [5] [6]. Falls have led to physical injuries, psychological distress, social isolation, and economic costs [7], making them the second leading cause of accidental injury deaths worldwide among the elderly [1]. Thus, prompt detection has been critical to reduce harm [8].

Wearable sensors have limitations such as discomfort and compliance issues [9]. So, vision-based methods have gained interest in the field of fall detection. Advances in pose estimation and deep learning have enabled the tracking of human joints to detect falls [9]. Combining object detectors like You Only Look Once (YOLO) with pose estimators such as AlphaPose has shown improved accuracy. On the other hand, models like ST-GCN have analyzed motion patterns to distinguish falls from other activities [9] [10].

This work proposes a novel method that leverages these advances to design real-time vision-based fall detection system in order to reduce false alarms and enables rapid alerts. The key contributions of this work are:

- Development and implementation of fall detection system that combines YOLO with AlphaPose and ST-GCN using spatial and postural fall logic.
- Reduction of false positives through context and tuning.
- Validation of the proposed system in diverse settings to ensure robustness.

The rest of the paper is organized as follows Section II reviews the relevant literature. Section III explains the devised system, including the datasets used and the modules developed that compromise the proposed system. Section IV presents the results obtained along with the evaluation and discussion. Finally, Section V concludes with findings and future research directions.

II. BACKGROUND

A. Vision-based methods:

Falls have remained a critical concern for the elderly, prompting the development of various detection systems aimed at ensuring timely intervention and enhancing safety. Researchers have extensively explored depth-camera-based solutions. Bian et al. [3], [11] have proposed a method using a single infrared depth sensor and Random Decision Trees (RDT), enabling joint localization even in low light, though it has struggled when individuals rest on furniture. Kwolek et al. [12] have improved accuracy by applying a ceiling-mounted depth camera and a k-Nearest Neighbors (KNN) classifier, maintaining privacy and robustness in poor lighting [13]. Vision-based fall detection approaches have generally been classified into three categories: single RGB camera, multi-camera, and depth-based systems. Researchers have preferred single RGB methods for their simplicity but have noted limitations due to occlusion and viewpoint sensitivity. Multi-camera setups have achieved viewpoint invariance but have required complex calibration. Depth-based systems, especially those using Kinect, have offered a practical balance between accuracy and implementation complexity. Techniques in this area have included thresholding, motion tracking, and skeleton modeling using classifiers such as SVM and ELM [14]. In pursuit of low-cost, real-time solutions, De Miguel et al. [15] have implemented a Raspberry Pi-based system that combines background subtraction and optical flow. Adhikari et al. [16] have trained CNNs on background-subtracted RGBD images,

while Nunez et al. [17] have used CNNs trained on optical flow for effective fall recognition across diverse datasets. Cippitelli et al. [18] have reviewed RGBD-based methods, identifying challenges and practical deployment trends. Alazrai et al. [6] have introduced the Motion-Pose Geometric Descriptor (MPGD) and used multi-class SVMs, achieving high recognition rates across various scenarios. Hossain et al. [19] have demonstrated reliable results using SVM classifiers on both depth videos and accelerometer data. Min et al. [20], [21] have addressed fall detection on furniture using Faster R-CNN to detect both people and objects in a scene. Vallabh et al. [22] and Nizam et al. [23] have developed adaptive systems that adjust sensitivity based on the individual's physical condition and risk profile. Harrou et al. [24] have proposed a GLR-SVM approach that reduces false positives by integrating statistical monitoring with classification. More recently, researchers have focused on skeleton-based models using pose estimation from RGB images. Alaoui et al. [25] have computed angles and distances between body keypoints and classified falls using traditional machine learning models. Chen et al. [26] have utilized OpenPose and symmetry principles to analyze body posture, achieving high performance. Lin et al. [27] have combined OpenPose with RNNs to reach 98.2% accuracy, while Ramírez et al. [28] have applied AlphaPose with KNN, achieving 98.84% accuracy and strong F1-scores on the UP-FALL dataset. Zheng et al. [10] have improved system efficiency by replacing YOLOv4's backbone with GhostNet and combining AlphaPose and ST-GCN for fall recognition. Salimi et al. [29] have achieved over 99% accuracy using TD-CNN-LSTM and 1D-CNN for pose-based fall detection. Beddiaar et al. [30] and Warke et al. [31] have presented comprehensive reviews, highlighting Kinect's advantages and limitations.

Alam et al. [9] have proposed a lightweight real-time fall detection system using Movenet, achieving high accuracy on low-compute devices. Mamchur et al. [32] have employed PoseNet, LSTM, and ensemble learning to detect falls with high precision. Gaya-Morey et al. [33] have reviewed 151 studies, emphasizing CNNs for spatial feature extraction, recurrent models for temporal patterns, and the importance of privacy-preserving methods such as skeleton-based data.

B. Sensor-based methods:

Several recent works have combined motion data and visual cues to improve fall detection. Kwolek and Kepski in [34] have integrated depth sensors and accelerometers to enhance detection accuracy using accelerometric thresholds. Yang et al. [35] have relied on geometric features from top-view Kinect depth images for real-time, threshold-based detection. Daher et al. [36] have fused smart floor sensors with accelerometers, using a hybrid feature selection method (HCM-SFS) to achieve over 90% sensitivity. Radar-based detection has also been explored by Jokanovic et al. [37], who have applied principal component analysis (PCA) to motion spectrograms, achieving 90% accuracy. Espinosa et al. [38] have developed a CNN-based multicamera system using optical flow features, outperforming traditional models. Reviews by Wang et al. [39]

and Palmerini et al. [40] have assessed sensor-based methods and multiphase model features, respectively, confirming strong performance of SVM-based models. Badgujar et al. [4] have used filtered accelerometer data to train classifiers, with Decision Tree achieving 95.87% accuracy. Nho et al. [41] have combined heart rate and motion data, validating a low-complexity, high-performance system. Finally, Vaiyapuri et al. [42] have introduced IMEFD-ODCNN, a smartphone-based deep learning model, showing notable improvements for smart eldercare.

C. Pose Estimation:

Human pose estimation has been a critical field in computer vision, with applications in healthcare, sports, and human-computer interaction. Deep learning, especially CNNs and transformers, has significantly improved accuracy and robustness in detecting body keypoints.

Pose as compositional tokens (PCT) approach in [43] has introduced a token-based representation of joint dependencies and has shown strong performance in occluded scenarios. Vision Transformer for Pose Estimation (ViTPose) has demonstrated that plain vision transformers can achieve state-of-the-art results with a simpler architecture, reaching 80.9% average precision (AP) on MS COCO [44]. OmniPose framework in [45] has advanced multi-person pose estimation with a waterfall module and multi-scale features, improving joint localization without post-processing. DECA approach introduced in [46] has tackled viewpoint equivariance using capsule networks and has achieved top-view accuracy in both depth and RGB domains. The proposed BUCTD method in [47] has combined bottom-up and top-down cues, leading to significant gains in crowded scenes like CrowdPose. For 3D pose from point clouds, Zhang et al. in [48] have proposed a method using density-guided sampling and temporal smoothing, which has improved stability and reduced noise. In [49], You Only Look Once, version 8, Single Pose (YOLOv8-SP) has been proposed to enhance posture detection for sports by integrating attention and polynomial corrections, showing higher joint angle accuracy. In [50], Context Coordinate Attention Module (CCAM)-Person has improved occlusion handling with attention modules, increasing precision attained when using MS COCO and CrowdPose. Fast Pose Distillation (FPD) method has addressed model efficiency, distilling knowledge from larger networks while retaining accuracy [51]. PoseRefiner method, which has been proposed in [52], has refined joint predictions in difficult cases using augmented data and a ResNet-based refinement step. YOLOv7 and YOLOv8 have integrated pose estimation modules alongside object detection, achieving high real-time accuracy and reliability in tasks like crowd monitoring and people counting [53]–[55].

III. PROPOSED FALL DETECTION SYSTEM

The proposed fall detection system is developed by enhancing an existing framework in [56] with advanced detection and action recognition methods. Initially, TinyYOLOv3 is used for person detection, AlphaPose for skeleton pose estimation, and

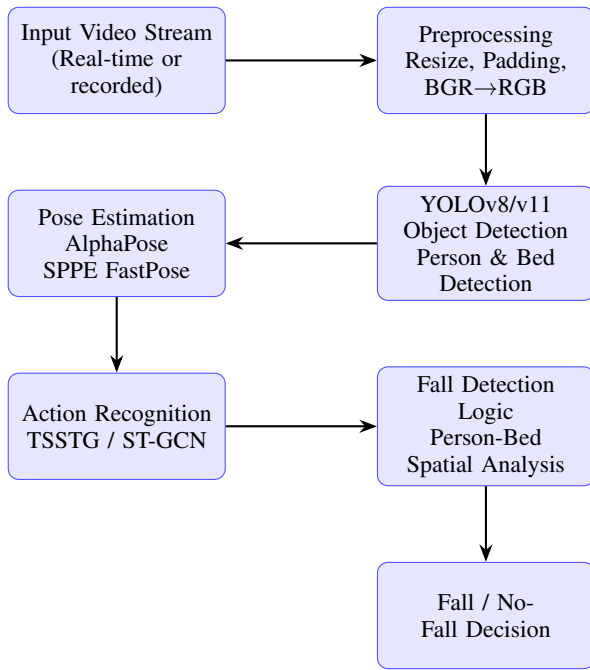


Fig. 1: Flowchart summarizing the fall detection pipeline using YOLOv8, AlphaPose, TSSTG, and fall detection logic based on spatial relationships.

the ST-GCN (TSSTG) model to classify seven actions from 30-frame sequences. To address the limited accuracy of the TinyYOLOv3 model, especially for small or complex objects, it is replaced by the more accurate and efficient YOLOv8 and YOLOv11 models, which also enable reliable bed detection. In addition, a novel fall detection logic is introduced to take advantage of the spatial relationship between the person and the bed, registering falls only when the person is detected below the surface of the bed. This context-aware approach significantly improves detection performance. The overall system workflow is illustrated in Fig. 1, which details the process starting from video input to fall decision.

A. Image Pre-processing and Post-processing

To ensure accurate and consistent detection, a streamlined pre-processing and post-processing pipeline is implemented. Pre-processing involves resizing the input images while preserving the aspect ratio and applying centered padding using the `ResizePadding` class, followed by the conversion of the color space from BGR to RGB to match the input format of the model. Post-processing applies Non-Maximum Suppression (NMS) to eliminate low-confidence and overlapping detections, resulting in clean, high-confidence bounding boxes that improved final detection accuracy.

B. Object Detection Models

In this work, two versions of YOLO models for object detection are used: YOLOv8 and YOLOv11. The following subsections explain briefly the models of both versions.

1) YOLOv8

YOLOv8 uses advanced deep learning with a modular design for fast and accurate detection. It features a CSP-Darknet backbone, multiscale fusion, and a detection head for objectness, classification, and bounding boxes. The pipeline includes aspect ratio resizing, single-pass inference, and NMS filtering, with real-time support via asynchronous processing and optimized loss functions.

2) YOLOv11

YOLOv11 uses a deeper backbone, attention mechanisms, enhanced feature fusion, and a multitask head predicting classes, boxes, and keypoints. It adds efficiency through separable convolutions, pruning, quantization, and hardware acceleration, enabling fast and precise detection on resource-limited devices.

C. Pose Estimation

AlphaPose utilizes the Single-Person Pose Estimation (SPPE) FastPose architecture with a ResNet50 backbone to improve feature extraction for accurate and efficient human pose estimation. ResNet50's deep convolutional structure and residual blocks enable capturing complex features while mitigating the vanishing gradient problem. The FastPose algorithm is optimized for real-time processing, combining ResNet50's powerful feature extraction with a streamlined prediction pipeline to detect 17 keypoints in the human body, covering the major joints and extremities. These keypoints form a skeletal structure and are used to analyze body posture and motion and are considered crucial to detecting dangerous or abnormal movements, such as falls.

D. Action Recognition

Action recognition utilizes deep learning to interpret complex human activities by extracting spatial and temporal features from video data. The TSSTG model, based on ST-GCN, processes human skeleton keypoints provided by AlphaPose, representing joints as graph nodes connected anatomically. It applies spatial convolutions within frames and temporal convolutions across frames to capture motion dynamics. The model's 9-layer architecture includes subsets for centripetal, centrifugal, and static motions, enhanced by ResNet blocks and dropout for robust training. The final classification is performed by a SoftMax layer that predicts the most likely action.

E. Fall Detection Logic

A custom fall detection logic complements action recognition by incorporating spatial context to reduce false positives, especially in bedroom environments. Upon TSSTG classification of "Fall Down" case, the bounding box of the person is analyzed for posture by comparing its width and height. A wider bounding box indicates a horizontal position typical of a fall. Additionally, all pose keypoints must lie below the vertical midpoint of the frame, indicating proximity to the ground. To avoid false positives caused by the cases where the person is lying on the bed, bed detection via YOLO models is used to

verify if the person is above or below the bed. Specifically, if the vertical midpoint of any bed bounding box falls within the upper 70% of the person's body height, the person is considered above the bed. A fall frame counter increments only when all these conditions are met consistently for more than six consecutive frames, ensuring temporal, spatial, and contextual reliability.

F. Used Datasets

In this work four datasets are used: GMDCSA [57], UMAFall [58], high quality fall simulation data [59], and KFall [60]. GMDCSA consists of webcam-recorded sequences of falls and ADLs performed by a single participant under consistent indoor lighting. UMAFall provides synchronized video and sensor data from 19 volunteers using smartphones. The high quality fall simulation data, which is provided by KU Leuven University, includes annotated fall scenarios recorded in indoor environments. The KFall dataset contains motion data from 32 young participants performing various ADLs and falls to support elderly fall detection research.

In this work, 79 videos from GMDCSA and 9 videos from UMAFall are used for training and validation; whereas, for testing 38 videos from GMDCSA, 14 videos from high quality fall simulation data, and 4 videos from KFall are used.

IV. RESULTS

A thorough evaluation of the proposed fall detection system is conducted to assess their performance, reliability, and readiness in real-time. The system is tested using the well known standard evaluation metrics, including accuracy, precision, recall, specificity, and F1-score, which are expressed in following equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

$$F1 - score = 2 \times \frac{P \times R}{P + R} \quad (5)$$

A. Evaluation and Discussion

Table 1 presents the achieved results. The original system based on TinyYOLOv3 model shows a strong precision (85.7%) and high specificity (94.6%), indicating that it is effective in minimizing false alarms. However, it suffers from extremely low recall (23.5%), which means that it fails to detect the majority of actual falls. Consequently, the F1-score is limited to 36.8%, indicating poor overall performance in real-world scenarios where missing a fall is critical.

The integration of YOLOv8 significantly improves the model's ability to detect fall events. The devised system

TABLE I: Evaluation results for the fall detection system

Metric	YOLOv11 + Fall Logic	YOLOv8 + Fall Logic	TinyYOLOv3 (Original)
Accuracy	73.9%	70.5%	53.4%
Precision	72.6%	68.6%	85.7%
Specificity	54.1%	43.2%	94.6%
Recall	88.2%	90.2%	23.5%
F1-Score	79.6%	78.0%	36.8%

based on YOLOv8 achieves a recall of 90.2%, highlighting its sensitivity to true positives. However, this improvement comes with a reduction in precision (68.6%) and specificity (43.2%), reflecting a higher number of false positives. Nevertheless, the F1-score increases substantially to 78.0%, showing a more balanced performance. The best performance is observed with the YOLOv11 model, which incorporates a refined fall detection logic based on human posture and spatial context. It achieves a precision of 72.6%, recall of 88.2%, and an F1-score of 79.6%. While its specificity (54.1%) is not matched that of the original model, it strikes a better balance between sensitivity and false positive control, making it more reliable for continuous monitoring applications.

These achieved results confirms that the combination of YOLOv11 and custom fall logic leads to the most balanced and effective fall detection solution, reducing missed detections while maintaining an acceptable rate of false alarms.

The three configurations (the original TinyYOLOv3 model, the updated YOLOv8-based model, and the enhanced YOLOv11-based system) tested on the same annotated videos containing a balanced number of fall and non-fall scenarios. The following figures demonstrate samples of the achieved test results from all used datasets. Fig. 2 shows a sample of fall detection on video from GMDCSA when the system does not compromise the devised fall logic. As noticed, the output shows a false alarm. Fig. 3 shows the system response on the same video when the fall logic is integrated. In addition, Fig. 4 and Fig. 5 demonstrate sample fall detections on videos from [59] and [58] respectively.

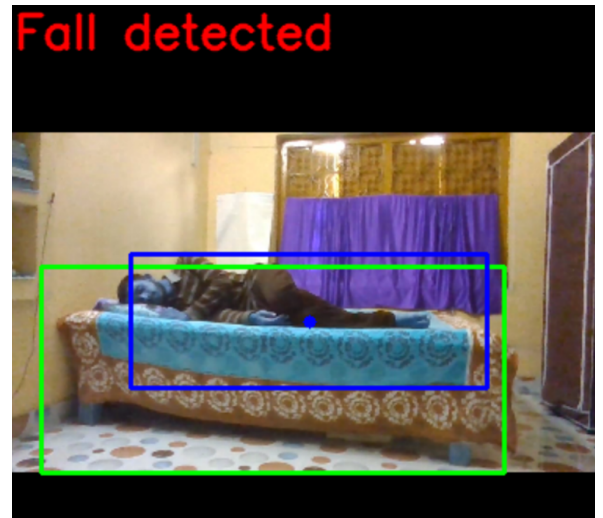


Fig. 2: Sample of fall detection on video from [57] without adopting the fall logic.

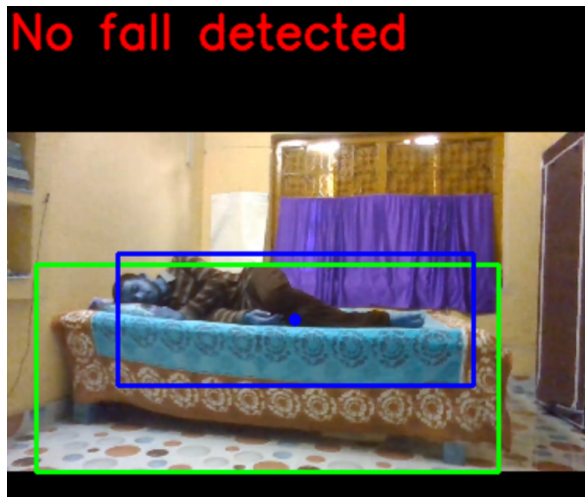


Fig. 3: Sample of fall detection on video from [57] when adopting the fall logic.

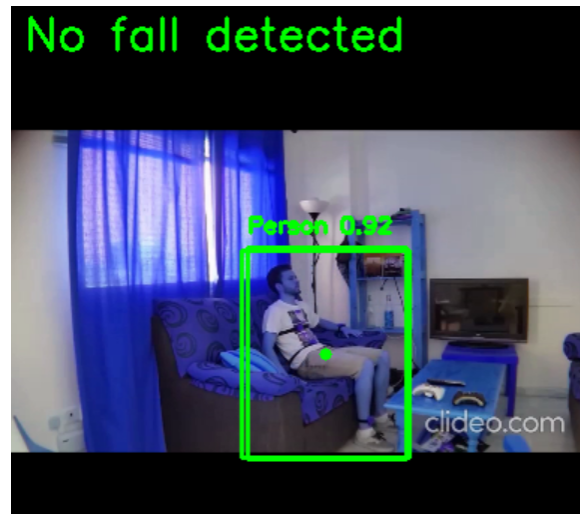


Fig. 5: Sample detection on video from [58]

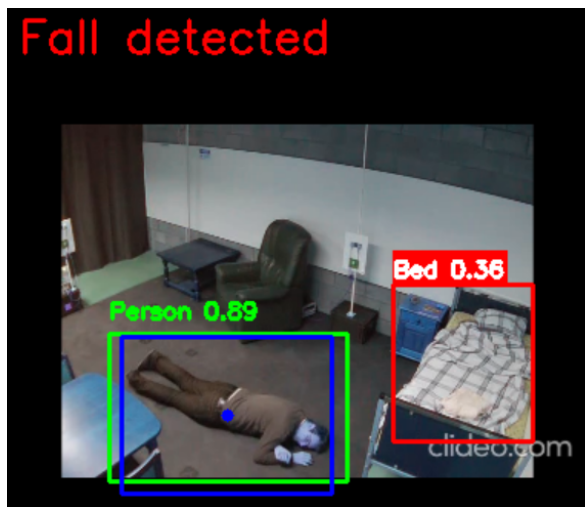


Fig. 4: Sample detection on video from [59]

V. CONCLUSION AND FUTURE WORK

In this paper, a novel fall detection system is developed and evaluated. The proposed system used TSSTG for action recognition and YOLO (v8 & v11) for person and bed detection with contextual fall logic to reduce false positives. The results achieved show that the proposed system improves precision, recall, F1-score, and reliability, showing strong potential for practical safety applications. Future work will focus on other models for object detection and pose estimation. In addition, future work will investigate the implementation targeting edge devices and the integration of real-time alert caregiver notifications into the developed system.

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